

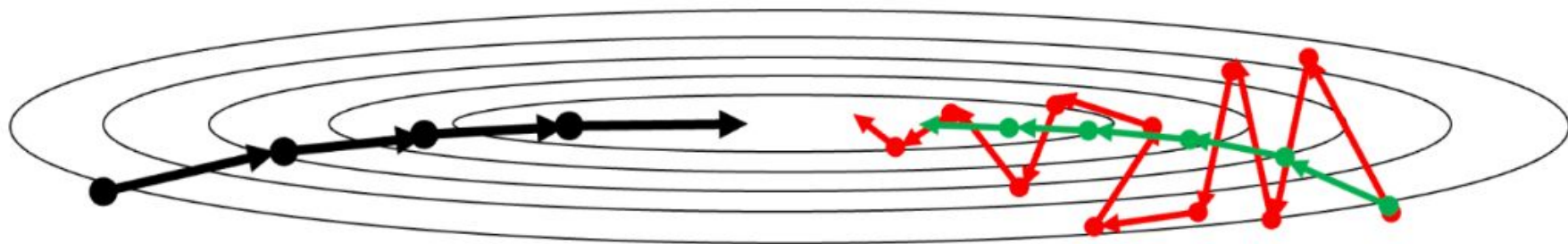
Introduction to Neural Networks

Seminar - Week 5 - Optimizers

Assignment 2 Goals

- Implement optimizers
 - SGD with Momentum
 - RMSProp
 - Adam

Gradient Descent - Momentum



Gradient Descent

$$W := W - \alpha dW$$

$$b := b - \alpha db$$

Gradient Descent with Momentum

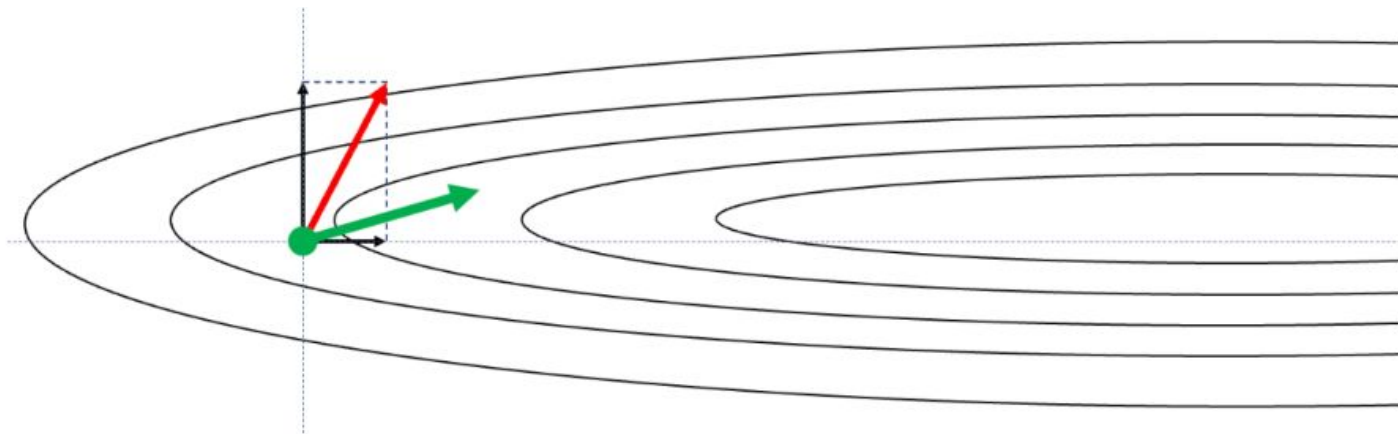
$$V_{dW} := (1 - \beta)dW + \beta V_{dW}$$

$$V_{db} := (1 - \beta)db + \beta V_{db}$$

$$W := W - \alpha V_{dW}$$

$$b := b - \alpha V_{db}$$

RMSProp



$$S_{dW} := (1 - \beta)dW^2 + \beta S_{dW}$$

$$S_{db} := (1 - \beta)db^2 + \beta S_{db}$$

$$W := W - \alpha \frac{dW}{\sqrt{S_{dW}}}$$

$$b := b - \alpha \frac{db}{\sqrt{S_{db}}}$$

Adam

$$V_{dW} := (1 - \beta_1)dW + \beta_1 V_{dW}$$

$$S_{dW} := (1 - \beta_2)dW^2 + \beta_2 S_{dW}$$

$$\widehat{V}_{dW} := \frac{V_{dW}}{1 - \beta_1^t}$$

$$\widehat{S}_{dW} := \frac{S_{dW}}{1 - \beta_2^t}$$

$$W := W - \alpha \frac{\widehat{V}_{dW}}{\sqrt{\widehat{S}_{dW} + \epsilon}}$$

Small value
for numerical stability

Week 5 - TASK

- Implement optimizers
 - SGD with Momentum
 - RMSProp
 - Adam

https://github.com/vgg-fiit/zneus_2025

Project 1 (9b)

- MLP

- Data analysis
- Data preprocessing and normalization
- Data split
- Configuration
- Experiment tracking
- Experiments - meaningful based on the results of previous experiments
 - Including improvement techniques (e.g. Dropout, Normalization layers, Skip Connections, Bottleneck Layers, ...)
- Hyperparameter search (grid/random) / sweep
- Results and evaluation metrics
- Clear code
- Markdown documentation and comments
- Final presentation of projects
- Effort on consultations

Project 1

- Consultation 1 - minimal requirements
 - Select team ([google sheet](#))
 - Select task (maximum 2 teams per task, first come, first serve)
 - Download and examine dataset
 - Exploratory data analysis ([example](#))
 - Select evaluation metrics
 - Check [THIS](#) and [THIS](#) video

Project 1 - List of topics

Credit Risk Assessment

- Binary Classification
- Dataset - <https://api.openml.org/d/31>

Speed Dating

- Binary Classification
- Dataset - <https://api.openml.org/d/40536>

Diabetes 130 US

- Binary Classification
- Dataset - <https://api.openml.org/d/45069>

Bioresponse

- Binary Classification
- Dataset - <https://api.openml.org/d/4134>

Magic Telescope

- Binary Classification
- Dataset - <https://www.openml.org/data/download/54003/MagicTelescope.arff>
<https://archive.ics.uci.edu/dataset/159/magic+gamma+telescope>
- paper - <https://publikationen.bibliothek.kit.edu/270043064/3813660>
 - paper about the creation of the data

Detroit

- Regression
- Dataset - <https://api.openml.org/d/552>

Houses

- Regression
- Dataset - <https://api.openml.org/d/537>

Covid 19

- Multiclass Classification
- Dataset - <https://www.kaggle.com/datasets/meirnazri/covid19-dataset>

Steel Plates Fault

- Multiclass Classification/Binary classification
- Bonus point - Utilize both approaches (either compare, or use together)
- Dataset - <https://api.openml.org/d/1504>